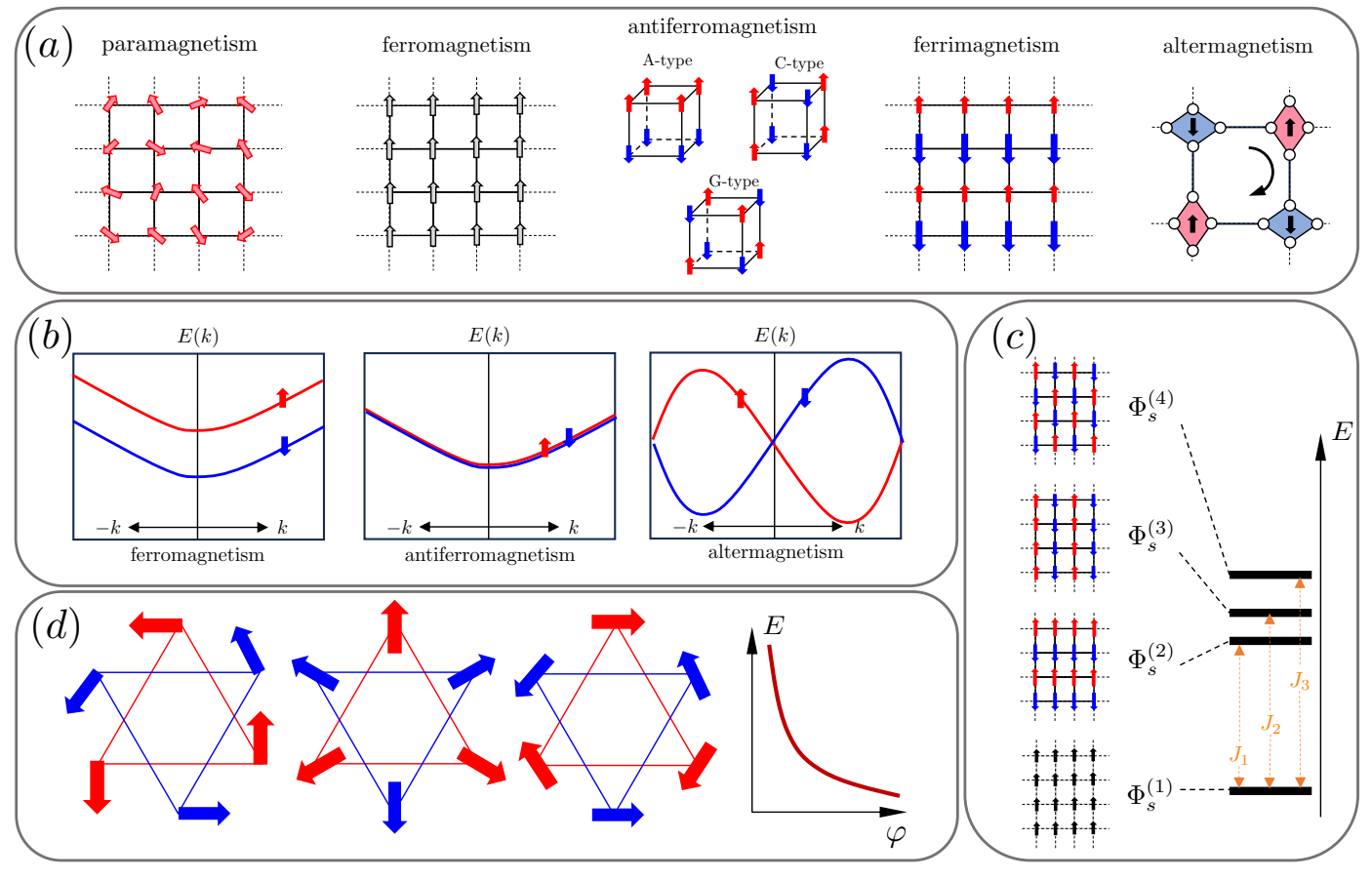

(a)
paramagnetism
ferromagnetism
antiferromagnetism
A-type
C-type
G-type
ferrimagnetism
altermagnetism
(b)
E(k)
−k
k
ferromagnetism
antiferromagnetism
altermagnetism
(c)
$\Phi_s^{(4)}$
$\Phi_s^{(3)}$
$\Phi_s^{(2)}$
$\Phi_s^{(1)}$
E
J_3
J_2
J_1
(d)
E
φ